

CS231 (Section A)

Surprise Quiz 4 (Bonus marks 0.3%)

Name:.....

Roll Number:.....

1. A system has only 4 CPUs, yet 20 instances of a.out are seen running (state "R") simultaneously in htop. Which OS concept explains this?

- A. Virtualization of the CPU
- B. Static linking
- C. Segmentation of the ELF file
- D. Protection rings

2. A process has finished its I/O and is waiting only to be assigned a CPU by the OS, with no other event pending. Which process state is it in?

- A. New
- B. Ready
- C. Waiting
- D. Terminated

3. Select all statements that are true about x86 protection rings as used by Unix-like operating systems?

- A. Level 0 is used for kernel mode / system calls
- B. Level 3 is used for user applications, the shell, and library functions
- C. Level 1 is used for device drivers
- D. All four levels (0-3) are used equally by user applications

4. A process attempts to divide a number by zero. As per Unix terminology, which kind of interrupt does this generate, and what is it called?

- A. A hardware interrupt, referred to as "trap"
- B. A software interrupt called a system call
- C. An exception
- D. A software interrupt, referred to as "trap"

5. Just before a libc wrapper (e.g., open()) executes the `int 0x80` instruction to trap into the kernel, which register must hold the unique system call number (as declared in unistd.h)?

- A. EAX
- B. EBP
- C. ESP
- D. EIP

6. Select all statements that are true about the IDT and the `int 0x80` system call path on x86?

- A. The IDT has 256 entries
- B. `int 0x80` causes the CPU to jump to the 128th IDT entry
- C. The 128th IDT entry points to the system call handler
- D. The IDTR register stores the system call number

Answer Key

1. A.
2. B.
3. A & B
4. C.
5. A.
6. A, B, and C.